Soil Erosion Risk with RUSLE

A 142 km² upland sub-watershed east of Malang (likely within Sub-DAS Amprong), East Java | Google Earth Engine + R | Self-initiated portfolio project using public data

The question

Where is soil loss highest in this watershed, and which land should conservation target first? The watershed runs from the lower valley near Malang up the western slopes of the Tengger massif, so it combines gentle farmland with very steep upper slopes.

Method

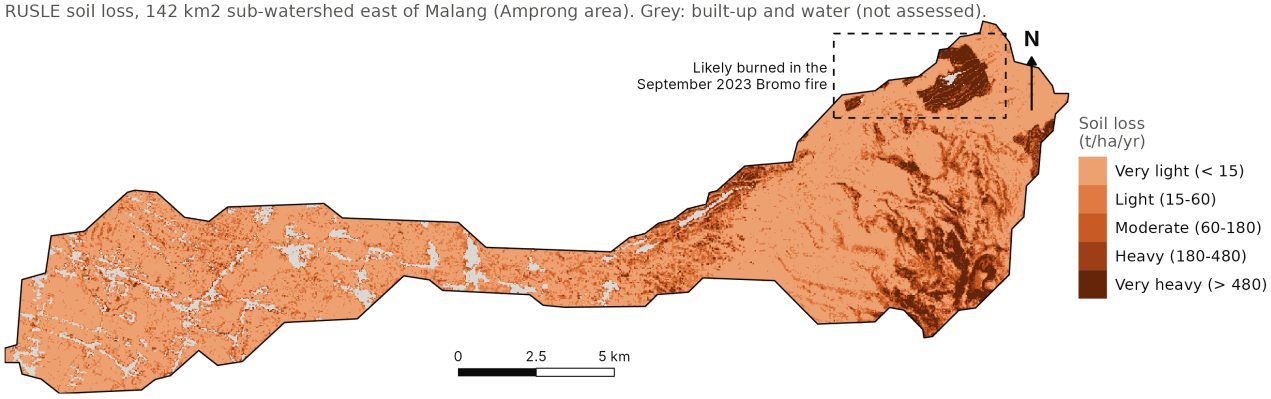
Factor	Data	Method
R rainfall erosivity	CHIRPS daily rainfall, mean 2014–2023	Renard & Freimund (1994)
K soil erodibility	OpenLandMap sand, clay, organic C (250 m)	Williams (1995), EPIC
LS slope length/steepness	Copernicus DEM 30 m; MERIT Hydro upstream area	Moore & Burch (1986), flow length ≤ 300 m
C cover management	Sentinel-2 NDVI 2023 (Cloud Score+)	Van der Knijff et al. (1999)
P support practice	ESA WorldCover 2021 cropland	Contouring by slope class; 1 elsewhere

A = R × K × LS × C × P at 30 m, built-up land and water excluded, grouped into the five erosion hazard classes used in Indonesia. Results were summarised in R by land cover and slope class.

Result 1: erosion hazard is concentrated on the steep upper slopes

Erosion hazard concentrates on the steep upper slopes

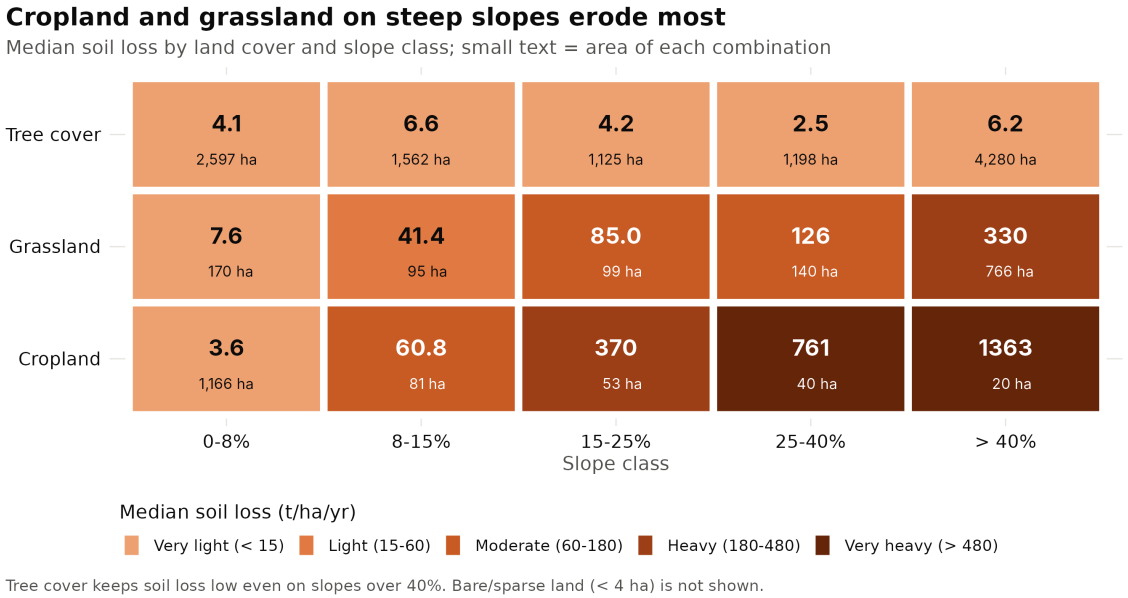
RUSLE soil loss, 142 km2 sub-watershed east of Malang (Amprong area). Grey: built-up and water (not assessed).



Classes as used for erosion hazard mapping in Indonesia. West = lower watershed towards Malang; east = Tengger massif.
Data: CHIRPS, OpenLandMap, Copernicus DEM GLO-30, MERIT Hydro, Sentinel-2, ESA WorldCover. Self-initiated portfolio project using public data.

Class (t/ha/yr)	Very light < 15	Light 15–60	Moderate 60–180	Heavy 180–480	Very heavy > 480
Area (ha)	8,889	2,131	1,055	651	670
Share	66.4%	15.9%	7.9%	4.9%	5.0%

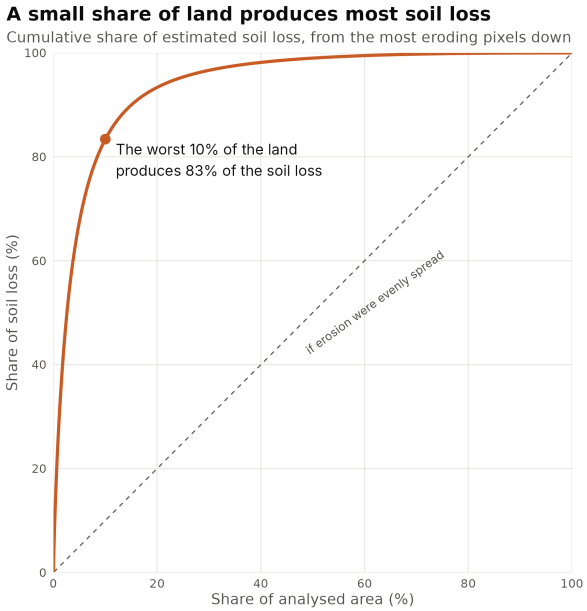
Result 2: land cover on steep slopes decides the outcome



Tree cover keeps soil loss low even on slopes over 40%. Bare/sparse land (< 4 ha) is not shown.

- **Grassland** covers 9.5% of the analysed area but produces **49% of the soil loss**; 60% of it lies on slopes over 40%.
- **Tree cover** keeps median soil loss below 7 t/ha/yr in every slope class, including slopes over 40%.
- **Cropland** is mostly on gentle slopes (86% on 0–8%, median 4.4 t/ha/yr). The 60 ha of cropland on slopes over 25% are among the most eroding land in the watershed.

Result 3: a small area produces most of the soil loss



The worst 10% of the land produces 83% of the estimated soil loss. The median is 5.7 t/ha/yr, while the mean (93 t/ha/yr) is pulled up by a small area of very steep, poorly covered land. **Priority areas** (cropland and grassland on slopes over 25% with heavy or very heavy erosion) cover **598 ha (4.5%)** and produce **48%** of the soil loss: revegetation, terracing or agroforestry there would address about half the problem on a twentieth of the land.

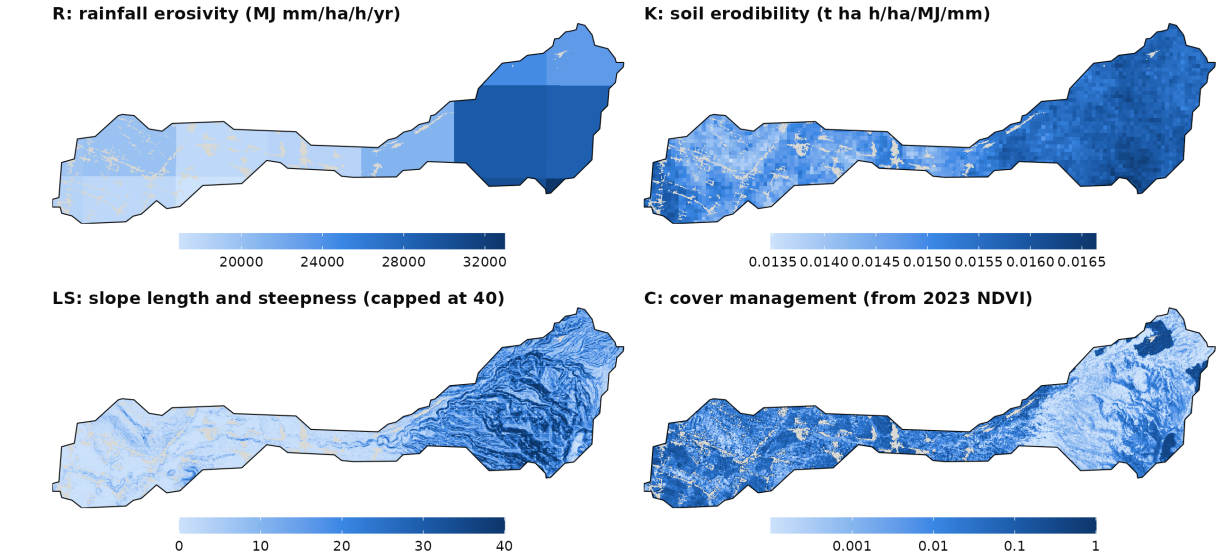
Result 4: a recent fire scar

A 270 ha patch near the caldera rim had very low vegetation cover in 2023 (median C 0.25) and was likely burned in the September 2023 Bromo savanna fire. It produces 30% of the estimated soil loss: a temporary post-fire peak, and the period when erosion control on burned slopes matters most. Without this patch the conclusions hold: the worst 10% of land still produces 81% of soil loss, grassland 53%, and priority areas (480 ha) 46%.

The factors behind the pattern

Slope and vegetation cover, not rainfall or soil, drive the pattern

RUSLE factor maps. K varies little at 250 m soil-map resolution; LS and C vary by orders of magnitude.



P (support practice) = 1 except on cropland, where contour farming values by slope class were applied (0.5-1.0).

Rainfall erosivity rises towards the Tengger slopes, and soil erodibility varies little at 250 m. Slope (LS) and vegetation cover (C) vary by orders of magnitude and drive most of the pattern.

Limitations

- RUSLE estimates gross hillslope soil loss, not sediment delivered to rivers; values are best used to rank areas.
- Slopes over 100% (5.4% of area) contribute 17.5% of the estimated loss; RUSLE was not developed for such slopes, so these values are likely overestimated.
- R uses an equation derived outside the tropics and ~5 km CHIRPS data; K comes from a 250 m global soil map.
- C is based on 2023 only, a dry El Niño year and the year of the Bromo fire; the burned patch was identified from the data and local knowledge, not an official fire perimeter. WorldCover 'tree cover' can include orchards and agroforestry.
- Results were not validated with field measurements; the HydroBASINS boundary is approximate.

What a client receives in this kind of project

- A commented Earth Engine script that computes every factor and can be re-run for any watershed.
- Factor maps, a soil loss map in hazard classes, and area statistics by land cover and slope.
- A priority-area analysis and a short report with assumptions and limitations stated clearly.

Data: CHIRPS (Funk et al. 2015); OpenLandMap (Hengl 2018, CC BY-SA 4.0); produced using Copernicus WorldDEM-30 © DLR e.V. 2010–2014 and © Airbus Defence and Space GmbH 2014–2018, provided under COPERNICUS by the EU and ESA, all rights reserved; MERIT Hydro (Yamazaki et al. 2019); modified Copernicus Sentinel data 2023; Cloud Score+; ESA WorldCover 2021 (CC BY 4.0); HydroBASINS. Software: Google Earth Engine; R 4.3.3 with terra, sf, ggplot2, patchwork.